

```

1.1.c -o tokenizer
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Skill/SKILL4$ ./tokenizer
Enter command: ls -l

--- TOKEN OUTPUT ---
Token 1: ls                Type: WORD
Token 2: -l                Type: WORD

--- VALIDATION ---
Validation: Token stream is valid
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Skill/SKILL4$ ./tokenizer
Enter command: ls -l /home

--- TOKEN OUTPUT ---
Token 1: ls                Type: WORD
Token 2: -l                Type: WORD
Token 3: /home             Type: WORD

--- VALIDATION ---
Validation: Token stream is valid
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Skill/SKILL4$ ./tokenizer
Enter command: ls -l | grep .c

--- TOKEN OUTPUT ---
Token 1: ls                Type: WORD
Token 2: -l                Type: WORD
Token 3: |                 Type: PIPE
Token 4: grep              Type: WORD
Token 5: .c                Type: WORD

--- VALIDATION ---
Validation: Token stream is valid

```

```

tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Skill/SKILL4$ ./tokenizer
Enter command: cat < input.txt

--- TOKEN OUTPUT ---
Token 1: cat               Type: WORD
Token 2: <                 Type: REDIRECT_IN
Token 3: input.txt         Type: WORD

--- VALIDATION ---
Validation: Token stream is valid

```

```

tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Skill/SKILL4$ ./tokenizer
Enter command: ls | | grep

--- TOKEN OUTPUT ---
Token 1: ls                Type: WORD
Token 2: |                 Type: PIPE
Token 3: |                 Type: PIPE
Token 4: grep              Type: WORD

--- VALIDATION ---
Error: Consecutive pipes are not allowed

```

```
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Skill/SKILL4$ gcc -Wall -g parser.c
-o parser
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Skill/SKILL4$ ./parser
Enter command: ls -l

--- PARSE TREE ---
COMMAND: ls -l

--- EXECUTION STRUCTURE ---
Execution type: COMMAND
Command: ls

Syntax validation: SUCCESS
```

```
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Skill/SKILL4$ ./parser
Enter command: ls -l | grep .c

--- PARSE TREE ---
PIPE |
  COMMAND: ls -l
  COMMAND: grep .c

--- EXECUTION STRUCTURE ---
Execution type: PIPELINE
Left command: ls
Right command: grep

Syntax validation: SUCCESS
```

```
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Skill/SKILL4$ ./parser
Enter command: ls | | grep
Syntax Error: Consecutive pipes
```

```
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Skill/SKILL4$ ./parser
Enter command: | ls
Syntax Error: Command cannot start with '|'
```

```
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Skill/SKILL4$ ./parser
Enter command:
Parser Error: Empty command
```